

Q. NO.	PART - A	STEP MARKS										
1	Define Bagging in ensemble learning. Bagging (Bootstrap Aggregating) is an ensemble technique where multiple models are trained on different random subsets of the training data, and their outputs are combined to improve accuracy and reduce overfitting.	2 Marks (Minimum 2 Point)										
2	Explain how Random Forest reduces variance in predictions. Random Forest reduces variance by creating many decision trees using random samples of data and random feature selection, then averaging their predictions to produce a more stable result.	2 Marks										
3	State the concept of K-means clustering. K-means clustering is an unsupervised learning algorithm that partitions data into <i>K</i> clusters by grouping similar data points around cluster centers called centroids.	2 Marks										
4	Describe how centroid is updated in clustering. In K-means clustering, the centroid is updated by calculating the mean position of all data points assigned to that cluster.	2 Marks										
5	Outline the purpose of Principal Component Analysis (PCA). PCA is used for dimensionality reduction by transforming high-dimensional data into fewer principal components while preserving maximum variance.	2 Marks										
6	Distinguish between hierarchical clustering and K-means. <table><thead><tr><th>Hierarchical Clustering</th><th>K-means Clustering</th></tr></thead><tbody><tr><td>Builds clusters in a tree-like structure</td><td>Divides data into fixed (K) clusters</td></tr><tr><td>Does not require predefined number of clusters initially</td><td>Requires predefined value of (K)</td></tr><tr><td>Computationally expensive</td><td>Faster and efficient for large datasets</td></tr><tr><td>Produces dendrogram output</td><td>Produces centroid-based clusters</td></tr></tbody></table>	Hierarchical Clustering	K-means Clustering	Builds clusters in a tree-like structure	Divides data into fixed (K) clusters	Does not require predefined number of clusters initially	Requires predefined value of (K)	Computationally expensive	Faster and efficient for large datasets	Produces dendrogram output	Produces centroid-based clusters	2 Marks
Hierarchical Clustering	K-means Clustering											
Builds clusters in a tree-like structure	Divides data into fixed (K) clusters											
Does not require predefined number of clusters initially	Requires predefined value of (K)											
Computationally expensive	Faster and efficient for large datasets											
Produces dendrogram output	Produces centroid-based clusters											